



Presentation

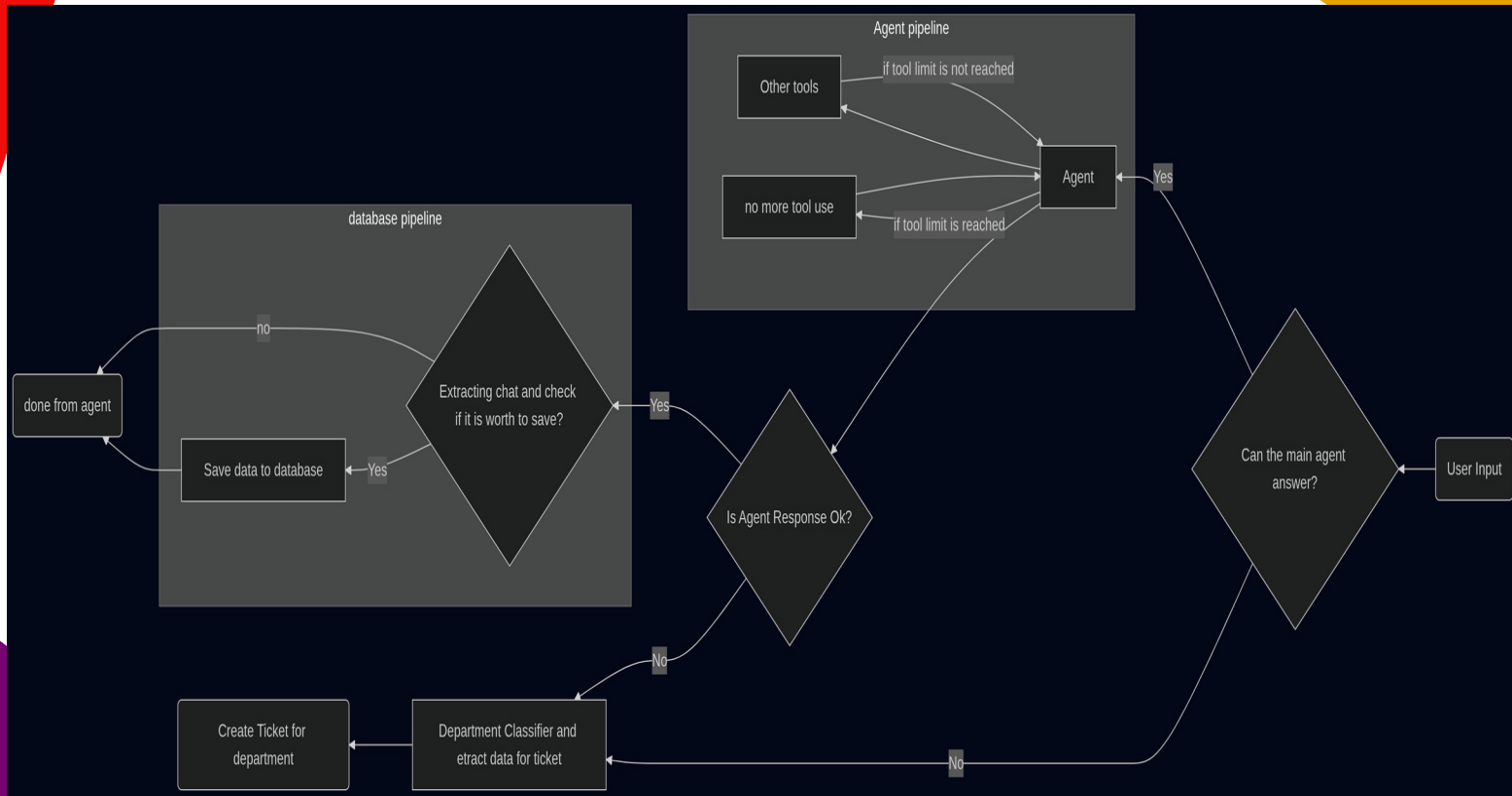
Support Agent



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Agent Core



Agent Core

- How agent uses PostgreSQL database?
- Agent uses PostgreSQL database as an AsyncSaver and AsyncStore.
- It can handle database by itself, because we pass it autocommit argument with True value.

Agent Core

- How does the agent use the knowledge base?
- It uses Google's "gemini-embedding-2" model.
- We pass the document (.txt or .md) to generate embeddings.
- The model returns them, and we save them to a database.

Agent Core

- How does the agent use the knowledge base? (continue)
- When the agent needs to retrieve information, it calls the retriever tool with the specified query text.
- The retriever receives the tool call and text, then sends the text to the embedding model to generate vectors.
- It performs a similarity search to find the most relevant data.
- Finally, the retriever returns the matched data as documents to the main agent.

Agent Core

- What to do for improve Agent Core?
- Switch to a more performant vector store like Pinecone instead of PostgreSQL.
- Use more advanced embedding models, such as OpenAI's embeddings.

Backend

- Our backend consists of 6 core features (entities):
 - 1) **Auth**: Handles user signup/signin, tokens, hashing, etc.
 - 2) **Chat Threads**: Manages user chats and messages within threads.
 - 3) **Feedback**: Handles user feedback (/).
 - 4) **Market Data**: Fetches and manages data from Yahoo Finance.
 - 5) **Tickets**: Manages support or service tickets.
 - 6) **Uploads**: Receives files from clients, processes them through the embedding model to generate vectors, and saves them to the database.

Backend

- What to do for improve Agent Core?
- There are so many things to do, like adding another route to remove uploaded files from vectorstore.
- Improving Auth feature.
- Etc ...

Frontend (client)

- We used prompt engineering and deepseek model to generate a client.
- We can improve this client with backend anytime.

Thank you!